



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2006

**HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #3**

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each section is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 72

- Attempt questions 1 – 6
- All questions are of equal value.

Examiner: *A.M. Gainford*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, x > 0$

Section A
(Start a new booklet.)

Question 1. (12 marks)

Marks

- (a) In how many ways can a committee of five be chosen from four women and six men, given that at least two must be women. 2
- (b) AB and AC are two chords of a circle on opposite sides of the centre, O . P and Q are the midpoints of AB and AC respectively. Prove that A , P , O and Q are concyclic. (Give reasons for each step.)
- (c) Use the substitution $u = 1 - x$ to find $\int_0^1 x\sqrt{1-x} \, dx$. 4
- (d) Write down the inverse function of $y = \sqrt{x-1}$ as a function of x and state its domain. 2
- (e) State the exact value of: 2
- (i) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$
- (ii) $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

Section continued overleaf.

Question 2 (12 marks)

Marks

6

(a) Differentiate (i) $\tan^{-1} 2x$.

(ii) $\sin^{-1}\left(\frac{x}{3}\right) + \cos^{-1}\left(\frac{x}{3}\right)$

(b) Find a primitive of:

6

(i) $\frac{1}{\sqrt{4-x^2}}$

(ii) $\frac{1}{1+4x^2}$

(iii) $\frac{1}{2x+1}$

End of Section A

Section B
(Start a new booklet.)

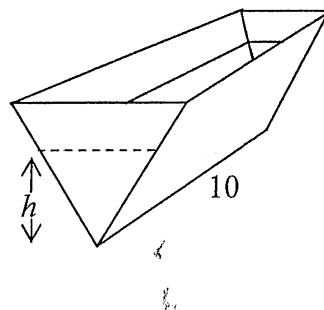
Question 3. (12 marks)

- | | | |
|-----|--|--------------|
| | | Marks |
| (a) | (i) State the domain and range of the function $f(x) = \sin^{-1}(-x)$. | 6 |
| | (ii) Sketch the graph of $y = f(x)$. | |
| | (iii) Find the gradient of the tangent to the curve at the point where it crosses the y -axis. | |
| (b) | Use the substitution $u = \log_e x$ to evaluate $\int_e^{e^2} \frac{1}{x \ln x} dx$. | 3 |
| (c) | A particle starts from O with acceleration $a = \cos t$ and velocity 0.5 m/sec . | 3 |
| | (i) Find its position when $t = 3\pi$ sec. | |
| | (ii) Find the total distance traveled in the first 3π seconds. | |
| | (iii) Sketch the graph of position against time for $0 \leq t \leq 3\pi$. | |

Question 4 (12 marks)

- | | | |
|-----|---|--------------|
| | | Marks |
| (a) | (i) Differentiate e^{-x^2} . | 6 |
| | (ii) Hence show that $\int_0^1 xe^{-x^2} dx = \frac{1}{2} \left(1 - \frac{1}{e} \right)$. | |

- | | | |
|-----|---|--------------|
| | | Marks |
| (b) | A large irrigation trough in the shape of a triangular prism with base angle 60° is being filled with water at a constant rate of $2 \text{ m}^3 / \text{min}$. | 6 |



- | | | |
|------|---|--|
| | | |
| (i) | Write an equation for the volume of water in the trough in terms of h , the depth of the water in metres. | |
| (ii) | Find the rate of change of depth of the water when the depth is 2 m . | |

End of Section B

Section C
(Start a new booklet.)

Question 5: (12 marks)

- | | Marks |
|--|--------------|
| <p>(a) The area bounded by the curve $y = \frac{b}{a}\sqrt{a^2 - x^2}$ (where a and b are constants), and the x-axis, is rotated about the x-axis.</p> <p>Find the volume of the solid of revolution so formed.</p> | 4 |
| <p>(b) Find the exact value of $\int_0^{\frac{\pi}{3}} \sin^2 x \, dx$.</p> | 4 |
| <p>(c) (i) Find the derivative of $x \ln x - x$.</p> <p>(ii) Hence evaluate $\int_1^2 \ln x^2 \, dx$.</p> | 4 |

Question 6: (12 marks)

- | | Marks |
|---|--------------|
| <p>(a) A rectangular table has eight seats, one on each end, and three on each side. Calculate the number of distinct ways that eight people may be seated around the table, if:</p> <p>(i) there are no restrictions,</p> <p>(ii) two particular people must not occupy end seats.</p> | 3 |
| <p>(b) Consider the curves $y = x^3$ and $y = \sqrt[3]{x}$.</p> <p>(i) Sketch the graphs of the two curves on the same axes, in the domain $0 \leq x \leq 1.5$.</p> <p>(ii) Determine the area of the region bounded by the curves in the stated domain.</p> | 3 |

Question continued overleaf.

Marks

(c) (i) Show that $x = 2.1$ is an approximate solution of the equation $x + \ln x = 3$.

3

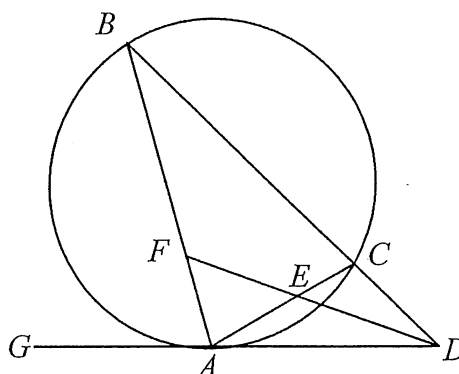
(ii) Use one application of Newton's Method to find a better solution.

(d) In the diagram at right AD is a tangent to the circle, and DF bisects $\angle ADB$.

3

(i) Copy the diagram to your answer sheet.

(ii) Prove that $AF = AE$.



End of Section C

This is the end of the paper.

Q (1) /12

Solution : Section A

4W	6M
2W	3M
3W	2M
4W	1M

[2]

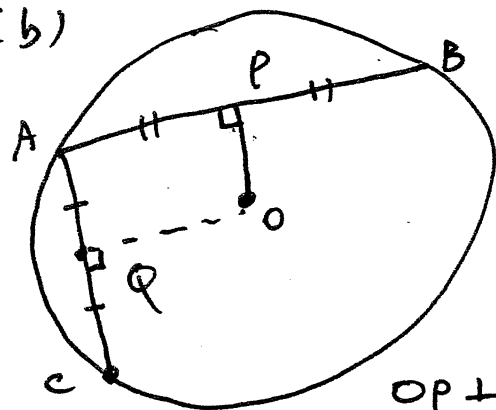
- At least 2W
- Choose 5 people.

∴ No. of ways

$$= \binom{4}{2} \binom{6}{3} + \binom{4}{3} \binom{6}{2} + \binom{4}{4} \binom{6}{1}$$

$$= 120 + 60 + 6 = 186.$$

(b)



OP ⊥ AB

- Line from centre
- of the circle to the mid pt of a chord is perp. to the chord.

Similarly, OQ ⊥ AC

∴ APOQ is a cyclic

- quad (opp angles are supplementary)

• i.e. A, P, O, Q are concyclic.

(c) Let $u = 1-x$, $du = -dx$

• When $x=0$, $u=1$, $x=1$, $u=0$

$$\therefore \int_0^1 x \sqrt{1-x} dx$$

$$= \int_1^0 (1-u) \sqrt{u} (-du)$$

$$= \int_0^1 (u^{\frac{1}{2}} - u^{\frac{3}{2}}) du$$

$$= \left[\frac{2}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} \right]_0^1$$

$$= \frac{4}{15} \quad [4]$$

(d) $f(x) = \sqrt{x-1}$ [2]

$d_f: x \geq 1, y \geq 0$

$$x = \sqrt{y-1} \therefore x^2 = y-1$$

$$\Rightarrow \boxed{f^{-1}(x) = x^2 + 1} \quad \begin{matrix} x \geq 0 \\ y \geq 1 \end{matrix}$$

(e) $\sin^{-1}(\frac{\sqrt{3}}{2}) = \frac{\pi}{3}$ [2]

(i) $\tan^{-1}(-\frac{1}{\sqrt{3}}) = -\tan^{-1}\frac{1}{\sqrt{3}}$

$= -\frac{\pi}{6}$ [1]

Q (2) /12

(i) $\frac{d}{dx} \tan^{-1}(2x)$

(a) $= \frac{2}{1+4x^2} \cdot \frac{1}{2(1+x^2)}$ [2]

(ii) $\frac{d}{dx} \left[(\sin^{-1} \frac{x}{3}) + (\cos^{-1} \frac{x}{3}) \right]$

$= \frac{1}{\sqrt{9-x^2}} - \frac{1}{\sqrt{9-x^2}}$

$= 0$ [4]

(b) (i) $\int \frac{dx}{\sqrt{4-x^2}} = \sin^{-1} \frac{x}{2} + C$ [2]

(ii) $\int \frac{dx}{1+4x^2} = \frac{1}{4} \int \frac{dx}{x^2 + \frac{1}{4}}$

$= \frac{1}{2} \tan^{-1}(2x) + C$ [2]

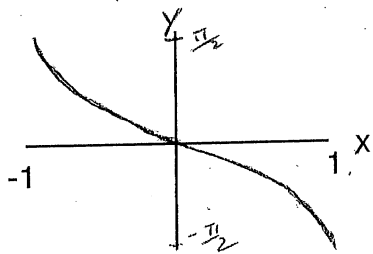
(iii) $\int \frac{dx}{2x+1} = \frac{1}{2} \int \frac{2dx}{2x+1}$

$= \frac{1}{2} \ln |2x+1| + C$ [2]

QUESTION 3

(a) (i) D: $-1 \leq x \leq 1$, R: $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(ii)



(iii) $y = \sin^{-1}(-x)$

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

$x = 0$, grad. of tangent = -1

(b) $u = \log x$, $\frac{du}{dx} = \frac{1}{x}$, $dx = x \cdot du$

when $x = e^2$, $u = 2$, when $x = e$, $u = 1$

$$\int_1^2 \frac{1}{x \cdot u} \cdot x \cdot du = \int_1^2 \frac{1}{u} du = [\ln u]_1^2$$

$$= \ln 2 - \ln 1 = \ln 2$$

(c) $a = \cos t = \frac{dv}{dt}$

$v = \sin t + c$

$t = 0$, $v = 0.5$, $c = 0.5$

$$v = \sin t + 0.5 = \frac{dx}{dt}$$

$x = -\cos t + 0.5t + k$

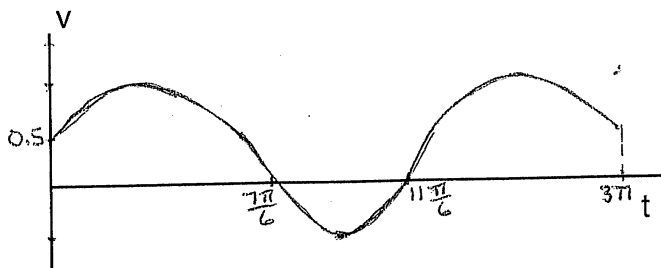
when $t = 0$, $x = 0$, $k = \cos 0 = 1$

$x = -\cos t + 0.5t + 1$

$t = 3\pi$, $x = \left(2 + 3\frac{\pi}{2}\right)m$

(d) (ii) distance travelled = $\int_0^{3\pi} (\sin t + 0.5) dt$

$v = \sin t + 0.5$



$$= 2 \left[0.5t - \cos t \right]_0^{7\pi/6} + \left[0.5t - \cos t \right]_{7\pi/6}^{3\pi}$$

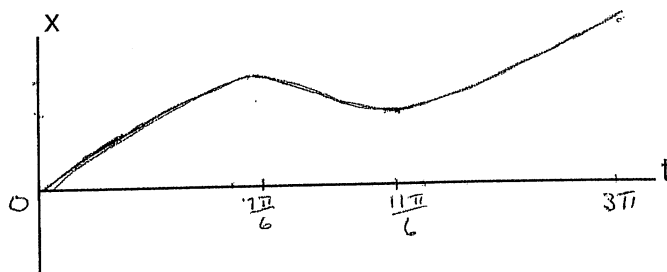
$$= 2 \left[\frac{7\pi}{12} + \frac{\sqrt{3}}{2} - (0 - 1) \right] + \left[\frac{11\pi}{12} - \frac{\sqrt{3}}{2} - \left(\frac{7\pi}{12} + \frac{\sqrt{3}}{2} \right) \right]$$

$$= \left(\frac{5\pi}{6} + 2\sqrt{3} + 2 \right) m$$

(iii) $\frac{dx}{dt} = \sin t + 0.5$

turning points $x = \frac{7\pi}{6}, \frac{11\pi}{6}$

$x = -\cos t + 0.5t + 1$



QUESTION 4

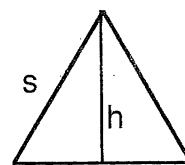
(a) (i) $\frac{d(e^{-x^2})}{dx} = -2xe^{-x^2}$

(ii) $\int_0^1 -2xe^{-x^2} dx = [e^{-x^2}]_0^1$

$$\int_0^1 xe^{-x^2} dx = -\frac{1}{2} [e^{-x^2}]_0^1$$

$$= \frac{1}{2} \left(1 - \frac{1}{e} \right)$$

(b) (i)



$$s = \frac{h}{\sin 60^\circ} = \frac{2h}{\sqrt{3}}$$

$v = \text{area of triangle} \times 10$

$$= \frac{1}{2} sh \times 10 = \frac{1}{2} \times \frac{2h}{\sqrt{3}} \times h \times 10$$

$$v = \frac{10h^2}{\sqrt{3}}$$

(ii) $\frac{dv}{dh} = \frac{20h}{\sqrt{3}}$, $\frac{dh}{dv} = \frac{\sqrt{3}}{20h}$

$$\frac{dh}{dt} = \frac{dh}{dv} \times \frac{dv}{dt}$$

$$\frac{dv}{dt} = 2$$

$$\frac{dh}{dt} = \frac{\sqrt{3}}{20h} \times 2$$

When $h = 2$, the rate of change of depth is

$$\frac{\sqrt{3}}{20} \text{ m/min}$$

SECTION C

$$5) a) \quad y = \frac{b}{a} \sqrt{a^2 - x^2}$$

Find x -intercepts (let $y=0$)

$$\frac{b}{a} \sqrt{a^2 - x^2} = 0$$

$$\sqrt{a^2 - x^2} = 0$$

$$a^2 - x^2 = 0$$

$$x^2 = a^2$$

$$x = \pm a$$

$$\text{Let } f(x) = \frac{b}{a} \sqrt{a^2 - x^2}$$

$$f(-x) = \frac{b}{a} \sqrt{a^2 - (-x)^2}$$

$$= \frac{b}{a} \sqrt{a^2 - x^2}$$

$f(x) = f(-x) \therefore$ Even. (symmetrical about y -axis)

$$V = \pi \int_{-a}^a y^2 dx$$

$$V = \pi \int_{-a}^a \left(\frac{b}{a} \sqrt{a^2 - x^2} \right)^2 dx$$

$$V = 2\pi \frac{b^2}{a^2} \int_0^a (a^2 - x^2) dx$$

$$V = 2\pi \frac{b^2}{a^2} \left[a^2 x - \frac{x^3}{3} \right]_0^a$$

$$V = \frac{2\pi b^2}{a^2} \left[a^3 - \frac{a^3}{3} - 0 \right]$$

$$V = \frac{2\pi b^2}{a^2} \left[\frac{2a^3}{3} \right]$$

$$V = \frac{4\pi b^2 a}{3} \quad \text{cubic units}$$

$$b) \int_0^{\frac{\pi}{3}} \sin^2 x \, dx$$

$$= \int_0^{\frac{\pi}{3}} \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) dx$$

$$= \left[\frac{1}{2}x - \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left(\frac{\pi}{3} \right) - \frac{1}{4} \sin \frac{2\pi}{3} - (0)$$

$$= \frac{\pi}{6} - \frac{1}{4} \left(\frac{\sqrt{3}}{2} \right)$$

$$= \frac{\pi}{6} - \frac{\sqrt{3}}{8}$$

$$c) i) y = x \ln x - x$$

$$\frac{dy}{dx} = x \cdot \frac{1}{x} + 1 \cdot \ln x - 1$$

$$\frac{dy}{dx} = 1 + \ln x - 1$$

$$\frac{dy}{dx} = \ln x$$

$$ii) \int_1^2 \ln x^2 \, dx$$

$$= \int_1^2 2 \ln x \, dx$$

$$= 2 \int_1^2 \ln x \, dx$$

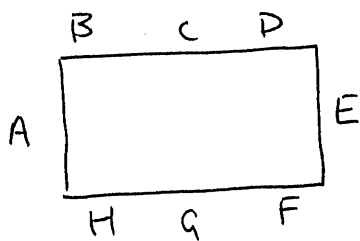
$$= 2 \left[x \ln x - x \right]_1^2$$

$$= 2 \left(2 \ln 2 - 2 - (1 \cdot \cancel{\ln 1} - 1) \right)$$

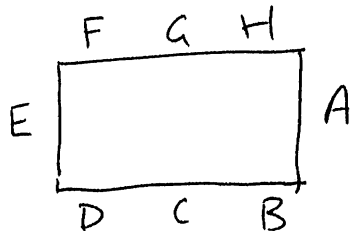
$$= 2 (2 \ln 2 - 1)$$

$$= 4 \ln 2 - 2$$

6) a) i)



would be the same as



so we must divide by 2

ie number of distinct ways is $\frac{8!}{2} = \frac{40320}{2}$
 $= 20160$

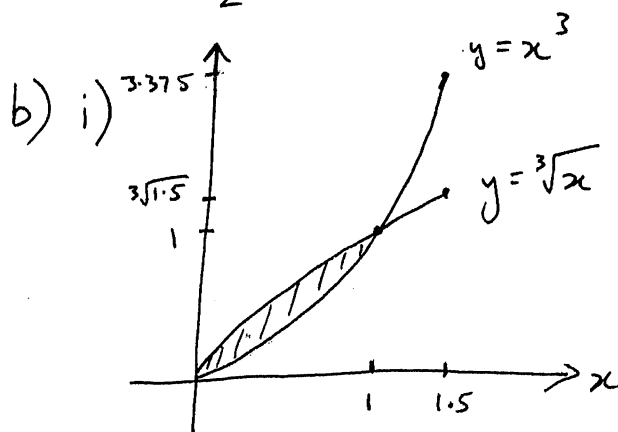
ii)

if two must not occupy the end seat

we are choosing 2 from 6 (not interested in order because of symmetry)

then choosing 6 from 6 (order matters)

$${}^6C_2 \times {}^6P_6 = 10800$$



ii) Area bounded in the domain (shaded)

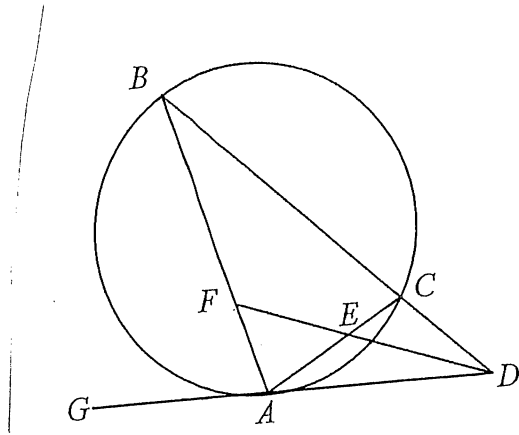
$$\begin{aligned} A &= \int_0^1 (x^{\frac{1}{3}} - x^3) dx \\ &= \left[\frac{3}{4} x^{\frac{4}{3}} - \frac{x^4}{4} \right]_0^1 \\ &= \left[\frac{3}{4} (1)^{\frac{4}{3}} - \frac{(1)^4}{4} \right] - [0] \end{aligned}$$

$$= \frac{3}{4} - \frac{1}{4}$$

$$= \frac{1}{2} \text{ square units}$$

c) (omitted)

d) i)



ii) let $\angle FDB = x$
 $\angle FDA = x$ (DF bisects $\angle ADB$)

let $\angle CAD = y$
 $\angle ABD = y$ (angle in the alternate segment equals the angle between the chord and tangent)

$\angle AFD = x + y$ (the exterior angle of $\triangle BFD$ is equal to the sum of the two opposite interior angles)

$\angle FEA = x + y$ (the exterior angle of $\triangle EAD$ is equal to the sum of the two opposite interior angles)

$\therefore \triangle AFE$ is isosceles ($\angle AFD = \angle FEA$)

$\therefore AF = AE$ (sides opposite equal angles in isosceles triangle).